

Weekly Report

iOS Application Development & SNCC

Date: May 12 – May 19, 2026
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Projects: Employ210 & SNCC

1. SNCC Project — Prompt Engineering Breakthrough

Significant progress was made on the SNCC behavioral classification system this week. Through systematic prompt engineering across four iterations, the classification accuracy was improved from **74% to 90%**. A detailed technical report documenting the methodology and findings has been prepared as a separate deliverable.

Accuracy Improvement: 74% → 90%

Key Achievements:

- **Confusion Matrix Analysis:** Identified three primary failure modes in the original prompt where the model was misclassifying behavioral categories. Each failure pointed back to specific language in the prompt definitions.
- **Five Targeted Edits:** Made precise, surgical changes to the prompt — refined category definitions, replaced broken heuristics with contextual rules, and added explicit boundary clarifications for A vs B, B vs C, and D vs C classifications.
- **Four Iteration Cycles:** Each version targeted the largest remaining error class from the previous round, resulting in measurable accuracy gains at each stage.
- **Detailed Report Delivered:** A comprehensive HTML report documenting the full methodology, including category definitions, the six guiding principles distilled from failures, and the complete v4 prompt as currently deployed in production. The report covers anatomy of each edit and lessons learned.

2. Employ210 — Stress Test Findings & Optimization Plan

Conducted stress testing on the Employ210 iOS application. The primary issue identified by users was the time taken for HTA generation, with average completion times ranging from **180 to 200 seconds**. A detailed optimization report was prepared outlining six prioritized solutions.

Proposed Speed Optimizations:

Solution	Time Saved
Switch to faster model (GPT-4o-mini / Claude Haiku)	~145 seconds

Streaming response (real-time output)	~177 seconds perceived
Pre-cached common tasks	~199 seconds
Reduce maximum tokens per step	~60–80 seconds
Single phase generation (later)	~90 seconds
Lambda backend optimization (later)	~30–50 seconds

3. Employ210 — New Features Implemented

Premium Splash Screen:

Designed and implemented a polished entry splash screen for the application. The screen features animated logo elements, gradient effects, and smooth transitions to provide a professional first impression for new users.

Video Upload Feature:

Added the ability for trainers to upload existing videos from their device library in addition to recording new ones. Both recorded and uploaded videos are saved to AWS S3 with the same organized folder structure (trainer ID / trainee ID / recordings).

Saved Programs Improvements:

Implemented Katie's requested changes for the saved programs page: program cards now display the task analysis title instead of generic text, steps are collapsed by default with expandable sections, and a new Completed Sessions section displays evaluation history with completion percentages.

4. Infrastructure & Team Coordination

Worked through AWS infrastructure alignment issues this week. After coordinating with Yassine regarding region consistency, the application configuration was updated to remain on us-east-1 for compatibility with the existing Lambda pipeline. New TestFlight builds were pushed with the updated configuration.

Team Coordination:

- Coordinated with **Yassine** on AWS Lambda integration and S3 region configuration
- Met with **Katie** to discuss new feature requests and SNCC prompt validation cases
- Assisted **Zack and Yassine** with troubleshooting video upload issues on their devices

Upcoming Tasks

Priority	Task	Project
High	Implement faster model and streaming for HTA generation	Employ210
High	Pre-cache common task HTAs for instant delivery	Employ210

Medium	Continue monitoring SNCC accuracy on new video samples	SNCC
Medium	Coordinate with Yassine on backend Lambda optimization	Employ210

Attached Reports:

- SNCC Evaluation — Prompt Engineering Report (74% → 90%)
- HTA Generation Speed Optimization Report